

Embedded SQL (Self-Learning)

Q. What is Embedded SQL? Discuss how SQL statements interact with host programming languages using Host Variables and Cursors. Differentiate between Static and Dynamic SQL.

****Definition to Remember****

****Embed SQL in a host programming language****
****pre-compiled****

1. Why Embedded SQL?

SQL alone lacks general-purpose constructs (UI, file I/O, loops, networking). Embedding SQL in a host language allows full-featured applications to interact with databases.

2. Key Mechanisms

A. Host Variables

Variable

B. Cursors

A standard mismatch

DECLARE cursor -> Define the cursor + SELECT query

OPEN

3. Static SQL vs Dynamic SQL

| Feature | Static SQL | Dynamic SQL |

| :- | :-

****Must-Write Points (for 10 marks)****

1. Embed

Quick Recall

`Embed
DECLA